3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Airborne Deployment of the Aerosol Mass Spectrometer during the ACE-Asia Field Campaign

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ABSTRACT

The Aerosol Mass Spectrometer (AMS) has been recently developed to provide real-time quantitative measurement of aerosol size and chemical composition over the 50 nm to 5 micron diameter size range. The AMS was deployed inside the CIRPAS Twin Otter airplane during the ACE-Asia field campaign. This was the first aircraft deployment for this type of mass spectrometer. The AMS operated unattended, and collected data successfully during most Twin Otter flights (15 out of 19). The main challenges encountered in this deployement and the strategies used or planned to overcome them will be described. The ACE-Asia field experiment was designed to characterize how particles emitted in eastern Asia (China, Korea, and Japan) affect climate through both direct and indirect radiative forcing. The submicron aerosol was present in discrete layers about 0.5-1 km deep, separated by layers with much lower aerosol mass concentration. Sulfate was a major aerosol component during all flights, and its size distribution was remarkably constant days and atmospheric layers. Ammonium, nitrate, and organics were also observed in the aerosol.